

Appendix A. Geometry of Correlation Matrices under the Quotient Affine-Invariant Riemannian Metric

Brain functional connectivity matrices can be naturally represented as symmetric positive-definite (SPD) matrices, which have been widely studied in neuroimaging analysis [2]. Since the brain functional connectivity matrices considered in this study are correlation matrices with unit diagonal entries, they belong specifically to the correlation matrix manifold rather than the general SPD manifold. In this section, we briefly introduce the geometric structure of correlation matrices under the quotient affine-invariant Riemannian metric (Q-AIRM) [1].

We consider the space $Corr(d)$, which is defined as

$$Corr(d) = \{C \in \mathbb{R}^{d \times d} : C = C^\top, x^\top C x > 0, \forall x \neq 0, \text{Diag}(C) = \mathbf{1}\}, \quad (\text{A.1})$$

where $\mathbf{1}$ denotes the all-one vector. The space $Corr(d)$ consists of all $d \times d$ positive-definite correlation matrices, and its dimension is $d(d-1)/2$ since the diagonal elements are fixed to one.

The correlation matrix manifold can be regarded as a quotient space of the SPD manifold under positive diagonal scaling. Specifically,

$$Corr(d) \cong Sym^+(d) / \mathcal{D}_{++}^d, \quad (\text{A.2})$$

where $\mathcal{D}_{++}^d$ denotes the Lie group of all d -dimensional positive diagonal matrices. For a correlation matrix $C \in Corr(d)$, its equivalence class is defined as

$$[C] = \{DCD : D \in \mathcal{D}_{++}^d\} \subset Sym^+(d). \quad (\text{A.3})$$

That is, the set of SPD matrices sharing the same correlation structure as C . Based on this quotient formulation, the Q-AIRM distance between two correlation matrices $C_1, C_2 \in Corr(d)$ is defined by the minimum AIRM distance between C_1 and all representatives in the equivalence class of C_2 :

$$d_{\text{Q-AIRM}}^2(C_1, C_2) = \min_{D \in \mathcal{D}_{++}^d} d_{\text{AIRM}}^2(C_1, DC_2D). \quad (\text{A.4})$$

where $d_{\text{AIRM}}(\cdot, \cdot)$ denotes the affine-invariant Riemannian distance on $Sym^+(d)$, which is given by

$$d_{\text{AIRM}}^2(C_1, C_2) = \left\| \text{Log} \left(C_1^{-1/2} C_2 C_1^{-1/2} \right) \right\|_F^2. \quad (\text{A.5})$$

Therefore, the Q-AIRM distance can be equivalently expressed as

$$d_{\text{Q-AIRM}}^2(C_1, C_2) = \min_{D \in \mathcal{D}_{++}^d} \left\| \text{Log} \left(C_1^{-1/2} D C_2 D C_1^{-1/2} \right) \right\|_F^2. \quad (\text{A.6})$$

The minimization problem in the definition of the Q-AIRM distance determines the optimal diagonal scaling matrix of C_2 within its equivalence class. Specifically, we define

$$\tilde{D} = \underset{D \in \mathcal{D}_{++}^d}{\text{argmin}} d_{\text{AIRM}}^2(C_1, D C_2 D), \quad (\text{A.7})$$

and denote the corresponding representative as

$$\tilde{C}_2 = \tilde{D} C_2 \tilde{D}. \quad (\text{A.8})$$

Then, $\tilde{C}_2$ is the representative of the equivalence class $[C_2]$ that is closest to C_1 under the AIRM metric.

Under the Q-AIRM metric, the geodesic between two correlation matrices C_1 and C_2 can be obtained by first constructing the AIRM geodesic between C_1 and $\tilde{C}_2$ on $\text{Sym}^+(d)$ and then mapping it back to $\text{Corr}(d)$ through the correlation projection $\pi(\cdot)$. Specifically,

$$\gamma_{C_1, C_2}^{\text{QA}}(t) = \pi \left(C_1^{1/2} \text{Exp} \left(t \text{Log} \left(C_1^{-1/2} \tilde{C}_2 C_1^{-1/2} \right) \right) C_1^{1/2} \right), \quad t \in [0, 1]. \quad (\text{A.9})$$

where $\text{Exp}(\cdot)$ and $\text{Log}(\cdot)$ denote the matrix exponential and logarithm, respectively.

Based on the above geodesic formulation, the exponential map under the Q-AIRM metric $\exp^{\text{QA}}(C, X) : T_C \text{Corr}(d) \rightarrow \text{Corr}(d)$ can be further obtained as

$$\exp^{\text{QA}}(C, X) = \pi \left(C^{1/2} \text{Exp} \left(C^{-1/2} \text{hor}(X) C^{-1/2} \right) C^{1/2} \right), \quad (\text{A.10})$$

where $\text{hor}(X)$ denotes the projection of X onto the horizontal space of the quotient manifold, $\text{Exp}(\cdot)$ represents the matrix exponential, and $\pi(\cdot)$ denotes the correlation projection.

Conversely, the logarithmic map under the Q-AIRM metric maps a correlation matrix C_2 to the tangent space $T_{C_1} \text{Corr}(d)$. First, the AIRM logarithmic map $\log^{\text{A}}(C_1, \cdot)$ is applied to the optimal representative $\tilde{C}_2$, yielding

$$V = \log^{\text{QA}}(C_1, \tilde{C}_2) = C_1^{1/2} \text{Log} \left(C_1^{-1/2} \tilde{C}_2 C_1^{-1/2} \right) C_1^{1/2}. \quad (\text{A.11})$$

According to the relationship between the logarithmic map under Q-AIRM and the differential of the correlation projection map:

$$\log^{\text{QA}}(C_1, C_2) = d_{C_1} \pi(V). \quad (\text{A.12})$$

where $d_{C_1} \pi : T_{C_1} \text{Sym}^+(d) \rightarrow T_{C_1} \text{Corr}(d)$ denotes the differential of the correlation projection π at C_1 . The correlation projection is defined as

$$\pi(\Sigma) = \text{Diag}(\Sigma)^{-1/2} \Sigma \text{Diag}(\Sigma)^{-1/2}, \quad \Sigma \in \text{Sym}^+(d). \quad (\text{A.13})$$

Since $C_1 \in \text{Corr}(d)$ and satisfies $\text{Diag}(C_1) = I_d$, we have

$$d_{C_1} \pi(V) = V - \frac{1}{2} (\text{Diag}(V)C_1 + C_1 \text{Diag}(V)). \quad (\text{A.14})$$

The derivation of the above formula is provided in Appendix B. Therefore, the Q-AIRM logarithmic map can be explicitly written as

$$\log^{\text{QA}}(C_1, C_2) = V - \frac{1}{2} (\text{Diag}(V)C_1 + C_1 \text{Diag}(V)). \quad (\text{A.15})$$

Appendix B. Derivation of the Differential of the Correlation Projection Mapping

We derive the differential in Eq. (A.14) below. Let the correlation projection mapping be defined as

$$\pi(\Sigma) = \Delta(\Sigma)^{-1} \Sigma \Delta(\Sigma)^{-1}, \quad \Delta(\Sigma) = \text{Diag}(\Sigma)^{1/2}.$$

To compute the differential $d_{C_1} \pi$ of π at $C_1 \in \text{Corr}(d)$, assume that the differential is well-defined. Consider the curve $\Sigma(t) = C_1 + tV$, where $\Sigma(0) = C_1$ and $\dot{\Sigma}(0) = V$. Then

$$\begin{aligned} d_{C_1} \pi(V) &= \left. \frac{d}{dt} \pi(\Sigma(t)) \right|_{t=0} \\ &= \left. \frac{d}{dt} \pi(C_1 + tV) \right|_{t=0} \\ &= \left. \frac{d}{dt} [\Delta(C_1 + tV)^{-1} (C_1 + tV) \Delta(C_1 + tV)^{-1}] \right|_{t=0} \\ &= \left. \frac{d}{dt} [\text{Diag}(C_1 + tV)^{-1/2} (C_1 + tV) \text{Diag}(C_1 + tV)^{-1/2}] \right|_{t=0}. \end{aligned}$$

Let $D = \text{Diag}(V)$. Since $C_1 \in \text{Corr}(d)$ satisfies $\text{Diag}(C_1) = I_d$, we have

$$\begin{aligned}
d_{C_1} \pi(V) &= \frac{d}{dt} \left[(I_d + tD)^{-1/2} (C_1 + tV) (I_d + tD)^{-1/2} \right] \Big|_{t=0} \\
&= \frac{d}{dt} \left[\left(I_d + \frac{t}{2} D + o(t) \right)^{-1} (C_1 + tV) \left(I_d + \frac{t}{2} D + o(t) \right)^{-1} \right] \Big|_{t=0} \\
&= \frac{d}{dt} \left[\left(I_d - \frac{t}{2} D \right) (C_1 + tV) \left(I_d - \frac{t}{2} D \right) + o(t) \right] \Big|_{t=0} \\
&= \frac{d}{dt} \left[C_1 + t \left(V - \frac{1}{2} (DC_1 + C_1 D) \right) + o(t) \right] \Big|_{t=0} \\
&= V - \frac{1}{2} (DC_1 + C_1 D) \\
&= V - \frac{1}{2} (\text{Diag}(V) C_1 + C_1 \text{Diag}(V)).
\end{aligned}$$

References

- [1] P. DAVID AND W. GU, *A Riemannian structure for correlation matrices*, Operators and Matrices, 13 (2019), pp. 607–627.
- [2] C. JU, R. KOBLER, A. COLLAS, M. KAWANABE, C. GUAN, AND B. THIRION, *Spd matrix learning for neuroimaging analysis: Perspectives, methods, and challenges*, IEEE Transactions on Pattern Analysis and Machine Intelligence, (2026).